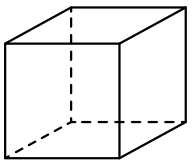
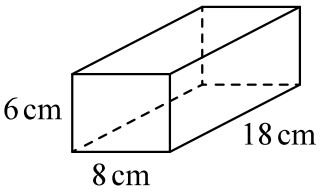
29 The diagram shows a cube and a cuboid.





The total surface area of the cube is equal to the total surface area of the cuboid.

Janet says,

"The volume of the cube is equal to the volume of the cuboid."

Is Janet correct?
You must show how you get your answer.

Cuboid surface area = $2(8\times18 + 8\times6 + 18\times6)$

$= 2(144 + 48 + 108) = 600\ cm^2$

Cube: $6s^2 = 600$, so $s^2 = 100$ and $s = 10$ cm

Cube volume = $10^3 = 1000\ cm^3$

Cuboid volume = $8 \times 18 \times 6 = 864\ cm^3$

No; the volumes are different.

5 marks total
Cuboid area 600
Cube edge 10 cm
Cube volume 1000
Cuboid volume 864
Janet is wrong

(Total for Question 29 is 5 marks)